

Dynamic Phi-Divergence Selection for Safe Exploitation in High-Dimension Leduc Hold'em

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Abstract

We study how the choice of ϕ -divergence used to regularize a response policy shapes the Exploitation-Exploitability tradeoff in Leduc Hold'em. We sweep six standard divergences across a grid of regularization strengths λ and trace the resulting Pareto frontiers. We then derive a theoretical envelope bounding the Exploitability gap for each of the divergences in the set and compare the results to the Pareto frontiers. To study adaptive divergence selection, we frame the (ϕ, λ) choice problem as a multi-armed bandit and solve it with Thompson Sampling under different levels of risk tolerance. Opponent uncertainty is modeled through a Beta-distributed mixture of passive and aggressive strategies that varies from round to round. We learn that the high-EV leaders of the static frontier (Jensen-Shannon, Squared Hellinger) perform well in EV-focused regimes, but are subpar when the player is risk-averse. This is because they accumulate too much Exploitability to survive the properties of their divergence penalty. On the other hand, the moderate performers of the static frontier (Forward KL, Chi-Squared) perform the best in these risk-averse environments because they drive Exploitability down the most.

1. Introduction

In a two-player zero-sum game, a player's expected return depends both on how well they exploit the opponent's mistakes and on how adaptive their strategy remains when they are uncertain of the opponent's model. Optimizing purely for expected value against an estimated opponent often yields specialized lines that earn high winnings in training but can be easily countered by an opponent who deviates from the assumed playstyle and can result in loss. The standard way to avoid this is to anchor the response policy to a Nash equilibrium, which guarantees a worst-case payoff equal to the value of the game, but typically forgoes substantial winnings against weak opponents. This phenomenon is known as the Exploitation-Exploitability tradeoff and has been studied extensively in the safe-exploitation literature ([Ganzfried and Sandholm, 2015](#)).

Comparatively less attention has been provided to how deviation from the safe baseline is measured. When a response policy differs from Nash Equilibrium, it does so by reallocating probability mass across actions at each information state. In this paper, we focus on how we can measure the distance between a response policy and its safe baseline. The ϕ -divergence from information theory offers a natural candidate space, with each generator inducing a different penalty from deviating from Nash equilibrium.

This paper conducts an analysis on Leduc Hold'em Poker, which has 936 information states. We make three contributions. First, we sweep the static Pareto frontier and find a sharp stratification in the best performing ϕ -divergences: Jensen-Shannon, Squared Hellinger, Total Variation, Forward KL, Reverse KL, and Chi-Squared. Second, to understand why certain divergences outperform others independently of factors like the optimizer and specific opponent model, we derive a theoretical envelope bounding the Exploitability gap by

$$\Delta E(\pi_i) \leq 2Bc_\phi\sqrt{L_i}\sqrt{D_\phi(\pi_i\|\pi_i^{\text{NE}})}.$$

The divergence-specific constant c_ϕ recovers the ordering of best-performing divergences on the Pareto frontier, while the $\sqrt{L_i}$ prefactor highlights how the high-dimensionality of the game impacts the Exploitability of our policies. Lastly, we run a multi-armed Thompson-Sampling bandit on Leduc and find sharper results on the optimal arm for each regime.

2. Preliminaries

To evaluate the tradeoff between exploiting a suboptimal opponent and remaining resilient to model misspecification, we must measure how far a proposed strategy deviates from a safe baseline. We frame this as a problem of evaluating distances between probability measures. Let $(\Omega, \mathcal{F})$ be a measurable space, where Ω is an arbitrary sample space and $\mathcal{F}$ is a σ -algebra over Ω . We consider two probability measures $P, Q \in \mathcal{P}(\Omega, \mathcal{F})$ ¹, where Q represents a fixed, unexploitable reference measure and P represents a candidate exploitative strategy. Assuming P is absolutely continuous with respect to Q ($P \ll Q$), the *Radon-Nikodym theorem* guarantees the existence of a non-negative, $\mathcal{F}$ -measurable derivative $\frac{dP}{dQ}$.

Definition 2.1. Let $\phi : (0, \infty) \rightarrow \mathbb{R}$ be a strictly convex function satisfying $\phi(1) = 0$. For $P, Q \in \mathcal{P}(\Omega, \mathcal{F})$ such that $P \ll Q$, the ϕ -divergence of P from Q is defined as:

$$D_\phi(P \parallel Q) := \int_\Omega \phi\left(\frac{dP}{dQ}\right) dQ = \mathbb{E}_Q \left[\phi\left(\frac{dP}{dQ}\right) \right]. \quad (2.1)$$

The geometric and statistical properties of this divergence are determined entirely by the generator ϕ . Rather than uniformly penalizing parameter differences, the choice of ϕ determines the structural asymmetry of the penalty. For instance, the logarithmic growth of Kullback-Leibler divergences greatly penalizes absolute continuity violations, e.g. assigning mass to actions the baseline strictly avoids, whereas the linear growth of Total Variation penalizes mass shifts uniformly across the support. These divergences and their corresponding generator functions are summarized in Table 1.

Divergence	Generator Function $\phi(t)$ ($t > 0$)
Forward Kullback-Leibler	$t \log t$
Reverse Kullback-Leibler	$-\log t$
Jensen-Shannon	$t \log\left(\frac{2t}{t+1}\right) + \log\left(\frac{2}{t+1}\right)$
Chi-Squared	$(t-1)^2$
Total Variation	$\frac{1}{2} t-1 $
Squared Hellinger	$(\sqrt{t}-1)^2$

Table 1: Summary of ϕ -divergences and their corresponding convex generator functions.

We embed these probabilistic concepts into the sequential decision-making framework of two-player, zero-sum extensive-form games (EFGs) with imperfect information. We use $i \in \{0, 1\}$ to denote the reference player, and $-i$ to denote their opponent. The game is formally defined by a tree of histories $\mathcal{H}$, where each history $h \in \mathcal{H}$ represents a sequence of actions of finite length taken from the root of the game. The leaves of this tree form a measurable subset of terminal histories $\mathcal{Z} \subset \mathcal{H}$ equipped with a σ -algebra $\mathcal{Z}$. At any non-terminal history $h \in \mathcal{H} \setminus \mathcal{Z}$, a player function $\rho(h) \in \{0, 1, c\}$ assigns the turn to either player 0, player 1, or a chance player c representing Nature, e.g. the randomness of the cards dealt to the players. In an imperfect-information game, a player does not perfectly observe the true underlying history h . To capture this mathematically, the histories assigned to player i are partitioned into a measurable space of information sets $(\mathcal{I}_i, \mathcal{I}_i)$. An information set $I \in \mathcal{I}_i$ is a collection of histories that player i cannot differentiate based on their observations. Because of this, the actions available to the player must be identical across the histories within I ; hence, we denote the localized action space as $\mathcal{A}(I)$, equipped with a σ -algebra $\mathcal{B}(I)$.

Definition 2.2. A *behavioral strategy* for player i is a Markov kernel that assigns a probability measure over the available actions to every information set. In particular, it is a mapping:

$$\pi_i : \mathcal{I}_i \rightarrow \bigcup_{I \in \mathcal{I}_i} \mathcal{P}(\mathcal{A}(I), \mathcal{B}(I)),$$

¹We denote by $\mathcal{P}(\Omega, \mathcal{F})$ the space of all valid probability measures defined on the measurable space $(\Omega, \mathcal{F})$.

such that $\pi_i(\cdot | I) \in \mathcal{P}(\mathcal{A}(I), \mathcal{B}(I))$ for all $I \in \mathcal{I}_i$, and for any measurable set $B \in \mathcal{B}(I)$, the map $I \mapsto \pi_i(B | I)$ is $\mathcal{I}_i$ -measurable. We denote the space of all valid behavioral strategies for player i as Σ_i .

A strategy profile $\pi = (\pi_i, \pi_{-i})$, coupled with a fixed stochastic probability measure for the chance player π_c , induces a probability measure $\mathbb{P}^\pi \in \mathcal{P}(\mathcal{Z}, \mathcal{Z})$ over the terminal histories, where $\mathbb{P}^\pi(z)$ represents the probability that terminal state $z \in \mathcal{Z}$ is reached following the strategy profile π . This induced measure captures the stochasticity of both the environment and the players' randomized choices. Let $U_i : \mathcal{Z} \rightarrow \mathbb{R}$ be an $\mathcal{Z}$ -measurable, bounded function that represents the payoff player i makes if the game ends at terminal state $z \in \mathcal{Z}$. The *expected utility* for player i is evaluated as the Lebesgue integral with respect to this induced measure:

$$u_i(\pi) := \int_{\mathcal{Z}} U_i(z) \mathbb{P}^\pi(dz) = \mathbb{E}_{z \sim \mathbb{P}^\pi} [U_i(z)].$$

By the zero-sum property, $u_0(\pi) = -u_1(\pi)$ for all profiles π .

Definition 2.3. A strategy profile $\pi^* = (\pi_i^*, \pi_{-i}^*)$ is a *Nash Equilibrium* if $u_i(\pi_i^*, \pi_{-i}^*) \geq u_i(\pi_i, \pi_{-i}^*)$ for all $\pi_i \in \Sigma_i$ and for both $i \in \{0, 1\}$.

Definition 2.4. The *Exploitability* of a strategy $\pi_i \in \Sigma_i$ is defined as

$$E(\pi_i) := \sup_{\pi_{-i}' \in \Sigma_{-i}} u_{-i}(\pi_i, \pi_{-i}'). \quad (2.2)$$

By von Neumann's minimax theorem, π_i^* satisfies a Nash equilibrium if and only if it minimizes E over Σ_i , and the minimum value equals the negative of the game's value for player i . Intuitively, $E(\pi_i)$ measures how punishable π_i is by an opponent that observes it and responds optimally; the Nash strategy π_i^* is the unique baseline that minimizes this punishability. Any deviation from π_i^* in pursuit of exploiting a weak opponent introduces additional Exploitability as the cost of that deviation, motivating the regularized objective in Section 3, where π_i^{NE} serves as the unexploitable reference measure Q to which any candidate response P is regularized.

3. Problem Formulation

Let $\pi_{-i}^{\text{opp}} \in \Sigma_{-i}$ be a fixed or estimated strategy profile for the opponent. The standard best-response objective seeks to maximize the reference player's expected utility $u_i(\pi_i, \pi_{-i}^{\text{opp}})$. To bound worst-case Exploitability, we restrict the optimization space by penalizing the distributional distance between the candidate strategy $\pi_i \in \Sigma_i$ and a pre-computed Nash equilibrium baseline π_i^{NE} . Because behavioral strategies act as Markov kernels, the divergence from π_i^{NE} is state-dependent. We combine these local divergences over the information set space $\mathcal{I}_i$ using a finite reference measure $\nu \in \mathcal{M}(\mathcal{I}_i, \mathcal{I}_i)$ ², which assigns a positive weight $w(I)$ to each information set. Given a strictly convex generator function ϕ , we define the *Regularized Response Problem* as:

$$\sup_{\pi_i \in \Sigma_i} \left[u_i(\pi_i, \pi_{-i}^{\text{opp}}) - \lambda \int_{\mathcal{I}_i} D_\phi(\pi_i(\cdot | I) \parallel \pi_i^{\text{NE}}(\cdot | I)) \nu(dI) \right]. \quad (3.1)$$

The parameter $\lambda \geq 0$ controls the regularization strength. As $\lambda \rightarrow 0$, the solution converges to the unconstrained best response. As $\lambda \rightarrow \infty$, the penalty dominates, forcing $\pi_i \rightarrow \pi_i^{\text{NE}}$, ν -almost everywhere, recovering the baseline. In standard imperfect-information games with finite state and action spaces, which includes any variation of Poker, we represent $\mathcal{I}_i$ and $\mathcal{B}(I)$ by the discrete power set σ -algebras generated by $\mathcal{I}_i$ and $\mathcal{A}(I)$ respectively. Furthermore, the information-set-weighting measure simplifies to a sum of weighted point masses, $\nu = \sum_{I \in \mathcal{I}_i} w(I) \delta_I$ ³. Any strategy measure π_i at set I is expressed as a linear combination of point masses over the action space:

$$\pi_i(A | I) = \sum_{a \in A} \pi_i(a | I), \quad \forall I \in \mathcal{I}_i, \forall A \in \mathcal{B}(I)$$

²Here, $\mathcal{M}(\mathcal{I}_i, \mathcal{I}_i)$ denotes the space of all finite measures on the measurable space $(\mathcal{I}_i, \mathcal{I}_i)$.

³Recall that the Dirac delta measure at a point x is defined for any measurable set A via the indicator function: $\delta_x(A) := \mathbb{1}_{x \in A}$.

Provided $\pi_i \ll \pi_i^{\text{NE}}$ at every $I \in \mathcal{I}_i$, both probability measures admit densities with respect to the counting measure. Thus, the Radon-Nikodym derivative evaluated at an action a is exactly the likelihood ratio of the point masses:

$$\frac{d\pi_i}{d\pi_i^{\text{NE}}}(a) = \frac{\pi_i(a \mid I)}{\pi_i^{\text{NE}}(a \mid I)}.$$

Using this we can discretize the integrand in (2.1) such that the Lebesgue integrals in (3.1) simplify into finite sums. The expected utility is computed exactly via a full traversal of the game tree, giving us the unconstrained optimization objective:

$$\max_{\pi_i \in \Sigma_i} \left[\sum_{z \in \mathcal{Z}} U_i(z) \cdot \mathbb{P}^{(\pi_i, \pi_{-i}^{\text{opp}})}(z) - \lambda \sum_{I \in \mathcal{I}_i} w(I) \sum_{a \in \mathcal{A}(I)} \pi_i^{\text{NE}}(a \mid I) \phi \left(\frac{\pi_i(a \mid I)}{\pi_i^{\text{NE}}(a \mid I)} \right) \right]. \quad (3.2)$$

4. Benchmarking ϕ -Divergence Regularization

The Regularized Response Problem stated in Objective (3.1) produces a distinct response policy for every (ϕ, λ) pair. To characterize the corresponding expected utility versus Exploitability Pareto frontiers, we perform an exhaustive sweep over the ϕ -divergence family and analyze how the choice of generator ϕ shapes the response trajectory.

4.1. Experimental Setup

We instantiate (3.2) in Leduc Hold'em, a two-player zero-sum imperfect-information game with a six-card deck (two suits, three ranks), two betting rounds separated by a public reveal, and 936 information sets for player 1. We parameterize $\pi_i(\cdot \mid I)$ via a per-information-set softmax over unconstrained logits $z(I) \in \mathbb{R}^{|\mathcal{A}(I)|}$ which removes the simplex constraints⁴ at the cost of non-convexity in z . The regularizer D_ϕ remains jointly convex in the action probabilities for every generator we consider. We optimize z with Adam (Kingma and Ba, 2015) for 300 steps at learning rate $\eta = 0.05$, computing exact expected values via tree traversal against a fixed evaluation opponent, π_{-i}^{opp} . We define π_{-i}^{opp} as a 50/50 per-state convex mixture of a passive strategy and an aggressive strategy⁵. We avoid evaluating against a purely passive opponent because it trivially folds to pressure, removing divergence-level differences by incentivizing the best response to just bet at every node.

4.2. Pareto Frontier Sweep

To trace the Pareto frontier for each generator ϕ , we solve (3.2) over 500 logarithmically-spaced regularization strengths $\lambda \in [10^{-5}, 10]$, producing 3,000 behavioral response policies. We score each strategy by a scalarized reward parameterized by a risk-aversion weight $\alpha \geq 0$:

$$R(\pi_i; \alpha) := \text{EV}(\pi_i) - \alpha \cdot E(\pi_i).$$

where $\text{EV}(\pi_i) := u_i(\pi_i, \pi_{-i}^{\text{opp}})$ is the expected utility of π_i against the fixed opponent we defined in Section 4.1. We fix $\alpha = 0.5$ throughout this section as a moderate tradeoff and defer the analysis of varying α to Section 5.

Divergence	EV @ $\lambda=10^{-5}$	E @ $\lambda=10^{-5}$	EV @ $\lambda=10$	E @ $\lambda=10$	Best R
Forward Kullback-Leibler	1.929	0.506	0.593	0.0035	1.676
Reverse Kullback-Leibler	1.929	0.506	0.593	0.0035	1.676
Jensen-Shannon	1.963	0.507	0.606	0.0089	1.710
Chi-Squared	1.743	0.343	0.591	0.0027	1.572
Total Variation	1.963	0.506	0.590	0.0030	1.710
Squared Hellinger	1.963	0.507	0.606	0.0089	1.710

Table 2: Per-divergence Pareto frontier extremes and best reward $R(\pi_i; 0.5)$

⁴The constraints are specified for each $I \in \mathcal{I}_i$ as $\pi_i(a \mid I) \geq 0, \forall a \in \mathcal{A}(I)$ and $\sum_{a \in \mathcal{A}(I)} \pi_i(a \mid I) = 1$

⁵The passive (or aggressive) strategy deterministically folds (or bets) whenever the action is legally permitted.

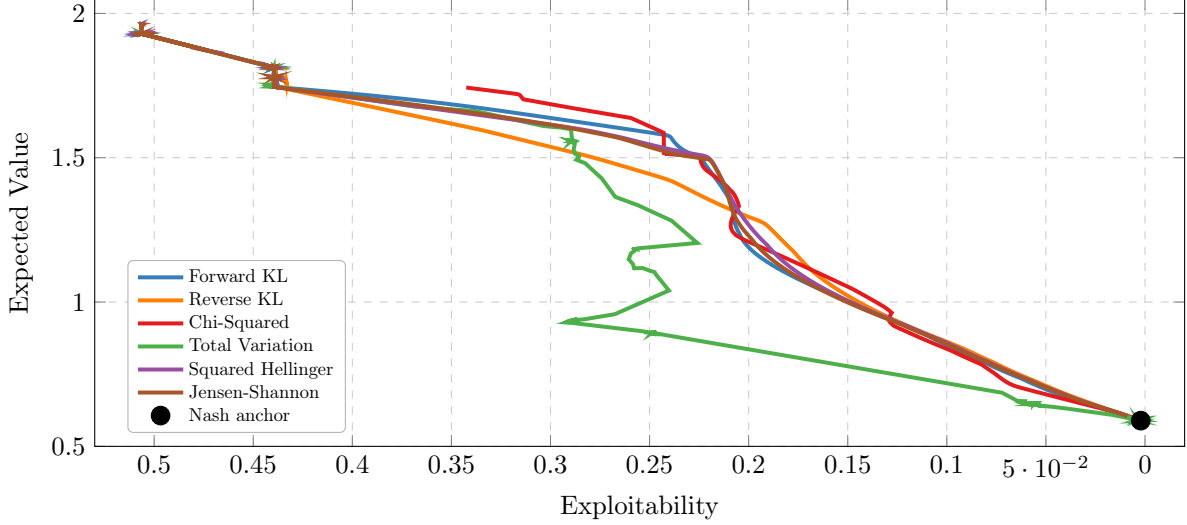


Figure 1: EV-versus-Exploitability Pareto frontiers for the six ϕ -divergences against the 50/50 Mixture opponent.

Figure 1 plots Expected Value against Exploitability for every (ϕ, λ) , with one curve per divergence. The upper-left of the plot corresponds to aggressive, exploitable play and the lower-right to Nash-aligned play; a divergence whose curve lies above another’s at equal Exploitability strictly dominates it. As the penalty vanishes, Jensen-Shannon, Total Variation, and Squared Hellinger reach the highest Expected Values ($EV \approx 1.96$), the Kullback-Leibler pair trails slightly ($EV \approx 1.93$), and Chi-Squared lags noticeably behind ($EV \approx 1.74$); its steep penalty prevents the policy from reaching the pure best response even at negligible λ . As λ grows, every divergence converges to the Nash baseline. Jensen-Shannon, Total Variation, and Squared Hellinger tie for the best reward at $R = 1.710$, the Kullback-Leibler pair follows at $R = 1.676$, and Chi-Squared trails at $R = 1.572$. Total Variation matches the leaders on reward, but its frontier is visibly non-monotone in λ ⁶. Across all six divergences the optimal λ sits near the grid floor of 10^{-5} , indicating that under moderate risk aversion the regularizer is most useful as a light correction.

4.3. Exploitability-Divergence Envelope

Figure 1 suggests that the choice of generator ϕ exerts a structural effect on the Exploitation-Exploitability tradeoff, beyond optimizer noise or the specific opponent realization. We make this claim precise by deriving an upper envelope on the Exploitability gap $\Delta E(\pi_i) := E(\pi_i) - E(\pi_i^{\text{NE}})$ in terms of the divergence penalty $D_\phi(\pi_i \| \pi_i^{\text{NE}})$ that the optimizer pays. The argument proceeds in three steps: (i) per-state variational inequalities place all six divergences on a common total-variation scale, isolating a constant c_ϕ ; (ii) Cauchy-Schwarz lifts the per-state bounds to the game tree, introducing a $\sqrt{L_i}$ prefactor; and (iii) the performance-difference lemma converts aggregate Total Variation drift into an Exploitability gap. The resulting envelope is loose by roughly three orders of magnitude in Leduc, but the qualitative ordering it predicts matches the empirical frontier ranking of Section 4.2 exactly.

4.3.1. Per-state variational rates

We first reduce all six divergences to a common scale. Total variation is the natural target because it directly measures probability-mass displacement without nonlinear reweighting, and because every other divergence in our family admits a Total Variation upper bound through standard variational inequalities.

Lemma 4.1. (*Sason and Verdú (2016)*) *Let p, q be probability vectors on a finite alphabet with $p \ll q$. Each ϕ -divergence in our family admits a total-variation bound in the form $\|p - q\|_{\text{TV}} \leq c_\phi \sqrt{D_\phi(p \| q)}$ with $c_{\chi^2} = \frac{1}{2}$, $c_{\text{KL}} = c_{\text{RevKL}} = \frac{1}{\sqrt{2}}$, $c_{\text{JS}} = c_{H^2} = \sqrt{2}$, and $c_{\text{TV}} = 1$.*

⁶The kink in $\phi_{\text{TV}}(t) = \frac{1}{2}|t - 1|$ at $t = 1$ is not handled well by Adam, producing numerous Pareto-dominated points.

4.3.2. Aggregating to the game tree

The regularizer in (3.2) aggregates per-state divergences across the player's information sets. Under uniform weighting $w(I) = 1$, we abbreviate the aggregate as

$$D_\phi(\pi_i \| \pi_i^{\text{NE}}) := \sum_{I \in \mathcal{I}_i} D_\phi(\pi_i(\cdot|I) \| \pi_i^{\text{NE}}(\cdot|I)).$$

The next lemma lifts the per-state Total Variation bound to this aggregate.

Lemma 4.2. *Fix any opponent strategy $\pi'_{-i} \in \Sigma_{-i}$. Let $\mathcal{Z}(I) \subset \mathcal{Z}$ be the set of terminal histories descending from I , and let $r(I) := \mathbb{P}^{(\pi_i^{\text{NE}}, \pi'_{-i})}(\mathcal{Z}(I))$ denote the reach probability of information set I under the joint profile $(\pi_i^{\text{NE}}, \pi'_{-i})$. Let $R := \sum_{I \in \mathcal{I}_i} r(I)$ be the expected number of player- i decision points visited per playthrough. For any $\pi_i \in \Sigma_i$ with $\pi_i \ll \pi_i^{\text{NE}}$ at every information set,*

$$\sum_{I \in \mathcal{I}_i} r(I) \|\pi_i(\cdot|I) - \pi_i^{\text{NE}}(\cdot|I)\|_{\text{TV}} \leq \sqrt{R} c_\phi \sqrt{D_\phi(\pi_i \| \pi_i^{\text{NE}})}.$$

Moreover $R \leq L_i$, where L_i is the maximum number of player- i decision nodes on any root-to-leaf path, so the bound also holds with $\sqrt{R}$ replaced by the constant $\sqrt{L_i}$.

Sketch. Write $r(I) \|\Delta_I\|_{\text{TV}} = \sqrt{r(I)} \cdot (\sqrt{r(I)} \|\Delta_I\|_{\text{TV}})$ with $\Delta_I := \pi_i(\cdot|I) - \pi_i^{\text{NE}}(\cdot|I)$. By Cauchy-Schwarz,

$$\sum_I r(I) \|\Delta_I\|_{\text{TV}} \leq \sqrt{\sum_I r(I)} \cdot \sqrt{\sum_I r(I) \|\Delta_I\|_{\text{TV}}^2}.$$

Applying Lemma 4.1 per information set and then $r(I) \leq 1$, the second factor satisfies $\sum_I r(I) \|\Delta_I\|_{\text{TV}}^2 \leq c_\phi^2 \sum_I r(I) D_\phi(\cdot|I) \leq c_\phi^2 D_\phi(\pi_i \| \pi_i^{\text{NE}})$. For $R \leq L_i$, each playthrough visits at most L_i player- i information sets, so $R = \sum_I r(I) = \mathbb{E}[\#\{\text{player-}i \text{ sets visited}\}] \leq L_i$. $\square$

4.3.3. The Exploitability-divergence envelope

The performance-difference lemma [Kakade and Langford \(2002\)](#) bounds how much a unilateral deviation from π_i^{NE} can increase the opponent's expected utility. In particular, this change is bounded by the deviating player's Total Variation drift at each information set, weighted by its reach probability. Substituting Lemma 4.2 into this result establishes our main envelope.

Theorem 4.3. *Let B bound the magnitude of any terminal payoff and let L_i be the maximum number of player- i decision nodes on any root-to-leaf path. For every square-root divergence in Lemma 4.1,*

$$\Delta E(\pi_i) \leq 2B c_\phi \sqrt{L_i} \sqrt{D_\phi(\pi_i \| \pi_i^{\text{NE}})}$$

sketch. Write $\Delta_I := \pi_i(\cdot|I) - \pi_i^{\text{NE}}(\cdot|I)$. Since the game is finite, let π_{-i}^* attain $E(\pi_i)$. Because $E(\pi_i^{\text{NE}})$ is a supremum over Σ_{-i} , it satisfies $E(\pi_i^{\text{NE}}) \geq u_{-i}(\pi_i^{\text{NE}}, \pi_{-i}^*)$, so the Exploitability gap collapses onto the single opponent π_{-i}^* :

$$\Delta E(\pi_i) \leq u_{-i}(\pi_i, \pi_{-i}^*) - u_{-i}(\pi_i^{\text{NE}}, \pi_{-i}^*).$$

The performance-difference lemma, taken with the reach probability under the baseline profile $(\pi_i^{\text{NE}}, \pi_{-i}^*)$ and the continuation value $Q := Q_{-i}^{(\pi_i, \pi_{-i}^*)}$, i.e. the expected future payout from an intermediate point in the game, under the deviated profile, expands this difference as $\sum_{I \in \mathcal{I}_i} r(I) \langle \Delta_I, Q(I, \cdot) \rangle$. Since terminal payoffs are bounded by B we have $\|Q(I, \cdot)\|_\infty \leq B$, and Hölder's inequality gives $|\langle \Delta_I, Q(I, \cdot) \rangle| \leq \|\Delta_I\|_1 \|Q(I, \cdot)\|_\infty \leq 2B \|\Delta_I\|_{\text{TV}}$. Therefore

$$\Delta E(\pi_i) \leq 2B \sum_{I \in \mathcal{I}_i} r(I) \|\Delta_I\|_{\text{TV}} \leq 2B c_\phi \sqrt{L_i} \sqrt{D_\phi(\pi_i \| \pi_i^{\text{NE}})},$$

where the final inequality applies Lemma 4.2 together with $R = \sum_I r(I) \leq L_i$. $\square$

For Leduc, the maximum terminal payoff is $B = 13$ chips, and the longest player- i action sequence has $L_i = 4$ decision nodes, so the envelope concretizes to

$$\Delta E(\pi_i) \leq 52c_\phi \cdot \sqrt{D_\phi(\pi_i \parallel \pi_i^{\text{NE}})},$$

with per-divergence envelope coefficients of 26 for Chi-Squared, 37 for Forward and Reverse KL, and 74 for Jensen-Shannon and Squared Hellinger.

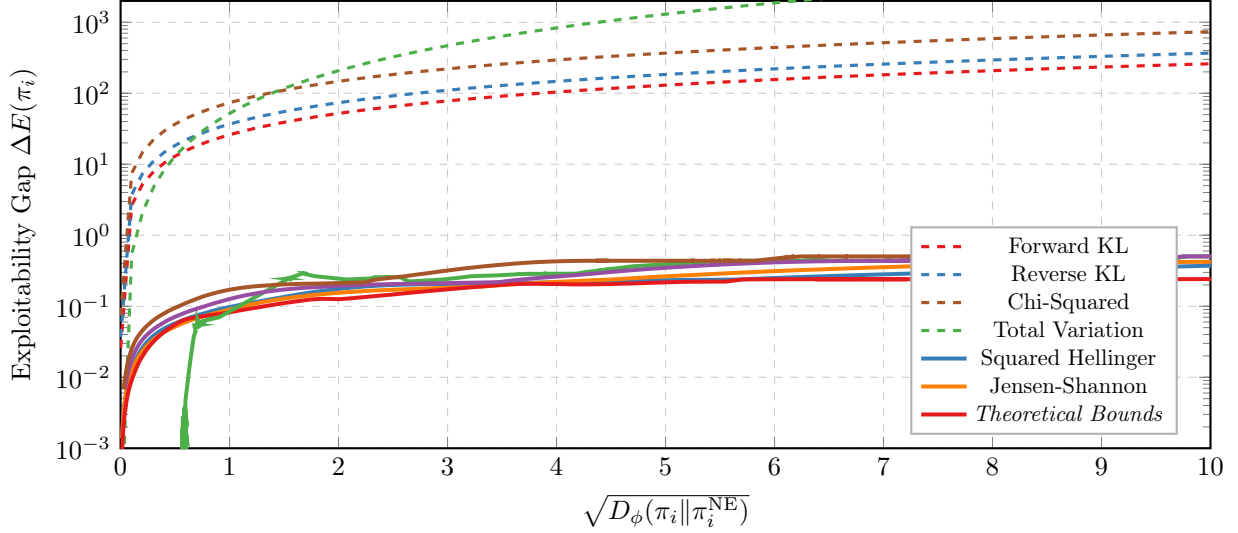


Figure 2: The empirical Exploitability gap $\Delta E(\pi_i)$ plotted against the normalized divergence penalty $\sqrt{D_\phi(\pi_i \parallel \pi_i^{\text{NE}})}$

5. Adaptive Divergence Selection through Multi-Armed Bandits

To determine which (ϕ, λ) pair a player should choose under opponent uncertainty, we frame the divergence selection problem as a finite-armed stochastic multi-armed bandit. Each arm corresponds to a candidate (ϕ, λ) pair, with rewards generated by evaluating the regularized objective against opponents sampled from a distribution of varying playstyles. We solve this bandit using Thompson Sampling, analyzing how the algorithm’s arm selection adapts to the player’s risk tolerance.

5.1. Bandit Formulation

Let $\mathcal{F} = \{\text{Forward KL, Reverse KL, Jensen-Shannon, Chi-Squared, Total Variation, Squared Hellinger}\}$ denote the six divergences from Table 1, and let $\Lambda = \{0.01, 0.25, 1.00, 5.00, 10.00\}$ be a logarithmically-spaced grid of regularization strengths. The arm space is the Cartesian product

$$\mathcal{A} = \mathcal{F} \times \Lambda, \quad |\mathcal{A}| = 30.$$

For each arm $a = (\phi, \lambda) \in \mathcal{A}$, we precompute the regularized response policy π_a^* by solving (3.2) with generator ϕ and strength λ , holding the training opponent fixed at the player’s prior expected model $\bar{\pi}_{-i}^{\text{opp}}$. This produces a fixed set of 30 candidate policies, one per arm. The bandit’s stochasticity comes from the opponent realization. Although each π_a^* is trained against $\bar{\pi}_{-i}^{\text{opp}}$, evaluation faces a parametric family $\{\pi_{-i}^{\text{opp}}(\theta)\}_{\theta \in \Theta}$ with $\theta_t \sim \mathcal{D}_\theta$ drawn independently each round. We calibrate the prior to the mean of the opponent distribution, $\mathbb{E}_{\theta \sim \mathcal{D}_\theta}[\pi_{-i}^{\text{opp}}(\theta)] = \bar{\pi}_{-i}^{\text{opp}}$, so the same arm produces different rewards on different rounds purely because of opponent variation around this mean.

On round t , the reward of arm a is the scalarized objective of Section 4.2 evaluated at the realized opponent:

$$R_a(\theta_t; \alpha) := u_i(\pi_a^*, \pi_{-i}^{\text{opp}}(\theta_t)) - \alpha \cdot E_a, \quad (5.1)$$

where $\alpha \geq 0$ is the player’s risk-aversion weight and $E_a := E(\pi_a^*)$ is the Exploitability of arm a ’s strategy, defined in (2.2). λ shapes which strategies are trained, while α scores them after training. Setting α large corresponds to low tolerance for exploitation, so arms near Nash should win. Only the first term in (5.1) is stochastic, since E_a is a deterministic property of the precomputed π_a^* ; all observed reward variance comes from opponent draws.

5.2. Thompson Sampling

We solve the bandit using Thompson Sampling (Russo et al., 2018) under a Gaussian-conjugate model. The algorithm maintains a normal posterior over each arm’s mean reward; on each round it draws one sample per arm, pulls the arm whose sample is largest, observes the realized reward, and updates that arm’s posterior. For each arm a we maintain a posterior $\mathcal{N}(\mu_a, \sigma_a^2)$ initialized at the non-informative prior $\mathcal{N}(\mu_0, \sigma_0^2) = \mathcal{N}(0, 1)$. Observed rewards are assumed corrupted by Gaussian noise $\mathcal{N}(0, \sigma_{\text{noise}}^2)$ with $\sigma_{\text{noise}} = 0.20$, calibrated to match the empirical reward variance under our Leduc opponent distribution. Gaussian-conjugacy gives the closed-form posterior update

$$\sigma_a^2 = \left(\frac{1}{\sigma_0^2} + \frac{n_a}{\sigma_{\text{noise}}^2} \right)^{-1}, \quad \mu_a = \sigma_a^2 \left(\frac{\mu_0}{\sigma_0^2} + \frac{S_a}{\sigma_{\text{noise}}^2} \right),$$

where n_a is the cumulative pull count of arm a and S_a its cumulative sum of observed rewards. After T rounds, the algorithm returns $\hat{a}^* := \arg \max_a \mu_a$ as its estimate of the optimal arm. The posterior-sampling mechanism trades exploration and exploitation automatically; arms with wide posteriors win the sample competition occasionally regardless of their mean estimates, while arms with high means and tight posteriors are pulled most of the time.

5.3. Experimental Setup

We instantiate the bandit in Leduc Poker. The opponent family $\{\pi_{-i}^{\text{opp}}(\theta)\}_{\theta \in [0,1]}$ interpolates between the passive strategy and the aggressive strategy, with θ being the probability of the passive action at every decision. To represent opponent variability across rounds, we sample $\theta_t \stackrel{\text{iid}}{\sim} \text{Beta}(2, 2)$, which concentrates most mass between 0.2 and 0.8 with mean 0.5. The training opponent of (3.2) corresponds to $\bar{\pi}_{-i}^{\text{opp}} = \pi_{-i}^{\text{opp}}(0.5)$, which is the 50/50 mixture opponent from Section 4.2, so the prior is calibrated to the mean of the realized opponent distribution.

Each bandit run consists of $T = 1000$ rounds. There is an initialization phase ($t = 1, \dots, 30$) in which every arm is pulled once to seed the posteriors, followed by 970 Thompson Sampling rounds. We run the full bandit with 20 independent random seeds; the opponent realizations $\{\theta_t\}_{t=1}^T$ are sampled once and held fixed across seeds, so cross-seed variation isolates exploration randomness from opponent randomness. We sweep the risk-aversion parameter over $\alpha \in \{0, 0.5, 2, 5, 10\}$. At each α we report the best-arm frequency and the average posterior mean and standard deviation per arm across seeds.

5.4. Results

The bandit results extend the static frontier of Section 4.2. Jensen-Shannon, Total Variation, and Squared Hellinger emerged there as the leading divergences, each with Best $R = 1.710$, followed by the Kullback-Leibler pair at 1.676 and Chi-Squared at 1.572. The bandit confirms this ranking at low α but increasingly diverges from it as the risk-aversion penalty rises, exposing three regimes.

α	Top-3 arms by avg. posterior mean	Mean range	Combined win freq
0.0	JS $\lambda=0.01$, Hell. $\lambda=0.01$, TV $\lambda=0.01$	[1.62, 1.69]	20/20
0.5	JS $\lambda=0.01$, Hell. $\lambda=0.01$, TV $\lambda=0.01$	[1.45, 1.51]	20/20
2.0	KL $\lambda=0.01$, χ^2 $\lambda=0.01$, TV $\lambda=0.01$	[0.95, 1.00]	17/20
5.0	χ^2 $\lambda=10$, χ^2 $\lambda=5$, KL $\lambda=10$	[0.524, 0.531]	12/20
10.0	χ^2 $\lambda=10$, χ^2 $\lambda=5$, KL $\lambda=10$	[0.514, 0.520]	18/20

Table 3: Top-3 arms by average posterior mean across 20 seeds. “Mean range” is $[\min, \max]$ of average posterior means within the top-3. “Combined win freq” is the number of seeds in which the bandit’s selected arm fell within the top-3 cluster.

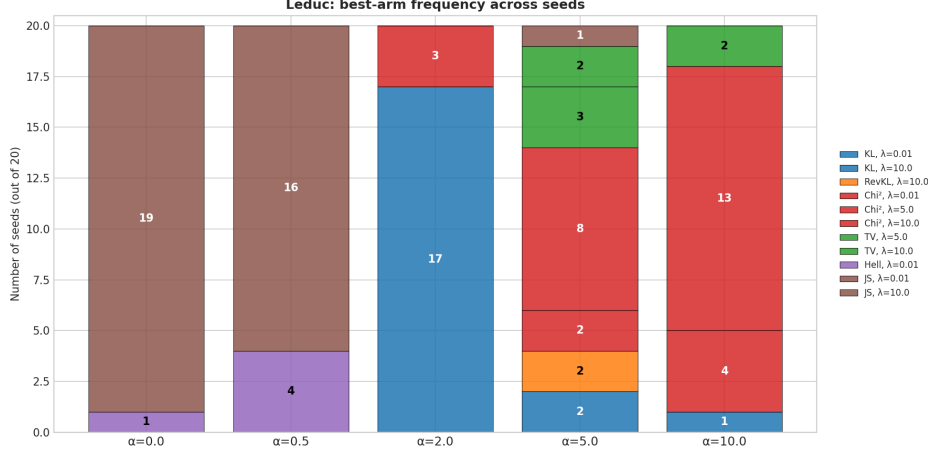


Figure 3: Best-arm frequency at each value of α across 20 seeds. Each stacked bar shows the distribution of selected arms.

EV-focused regime ($\alpha \leq 0.5$). At $\alpha = 0$, JS at $\lambda = 0.01$ wins 19/20 seeds with Squared Hellinger at the same λ taking the remainder. At $\alpha = 0.5$, the split is 16/20 to JS and 4/20 to Squared Hellinger. Jensen-Shannon and Squared Hellinger sit atop the static frontier, and once the Beta(2,2) opponent mixture is layered on, Jensen-Shannon’s slightly higher Expected Value at $\lambda = 0.01$ (1.724 versus 1.708) is enough to win posterior-sampling competitions under low Exploitability pressure.

Balanced regime ($\alpha = 2$). The bandit’s selection sharply changes: Forward KL at $\lambda = 0.01$ wins 17/20 seeds with the remaining 3/20 going to Chi-Squared at $\lambda = 0.01$. The reversal is captured by the deterministic reward proxy $EV - 2 \cdot E$ at $\lambda = 0.01$:

$$\begin{aligned}
\text{Forward KL} &: 1.493 - 2(0.223) = 1.047, \\
\text{Chi-Squared} &: 1.480 - 2(0.224) = 1.032, \\
\text{Total Variation} &: 1.662 - 2(0.338) = 0.986, \\
\text{Jensen-Shannon} &: 1.724 - 2(0.415) = 0.894.
\end{aligned}$$

The high-EV leaders accumulate roughly 50 to 100 percent more Exploitability than Forward KL and Chi-Squared and do not survive the doubled penalty. Notably the optimal λ has not shifted from the floor 0.01. This suggests that the regularizer is still useful only as a light correction, but the divergence paired with that light regularization switches from a loose-envelope generator (Jensen-Shannon) to tighter-envelope generators (Forward KL, Chi-Squared).

Safety-dominated regime ($\alpha \geq 5$). The bandit shifts toward strong regularization. At $\alpha = 5$, Chi-Squared at $\lambda = 10$ wins 8/20 seeds, with Chi-Squared at $\lambda = 5$ (2/20) and Total Variation at $\lambda = 5$ (3/20) following. At $\alpha = 10$, Chi-Squared at $\lambda = 10$ takes 13/20, then Chi-Squared at $\lambda = 5$ (4/20), Total Variation at $\lambda = 10$ (2/20), and KL at $\lambda = 10$ (1/20). The reward is now dominated by the Exploitability term, and arms that drive E_a closest to zero win. At $\lambda = 10$ the leading arm Exploitabilities order as

$$\begin{aligned}
\text{Chi-Squared} &: E_a = 0.0027, \\
\text{Total Variation} &: E_a = 0.0029, \\
\text{Forward KL} &: E_a = 0.0035, \\
\text{Reverse KL} &: E_a = 0.0035, \\
\text{Squared Hellinger} &: E_a = 0.0089, \\
\text{Jensen-Shannon} &: E_a = 0.0089.
\end{aligned}$$

Chi-Squared at $\lambda = 10$ attains the lowest Exploitability in the grid. Squared Hellinger and Jensen-Shannon, despite dominating at low α , never appear in the high- α winning set because their exploitabilities at $\lambda = 10$

are roughly three times higher than the Chi-Squared floor. The quadratic penalty of Chi-Squared, which under-performed in the static analysis precisely because it suppressed exploitation, becomes the dominant property once robustness rather than exploitation is the objective. This ordering matches the prediction of Theorem 4.3 exactly as Chi-Squared has the tightest envelope constant $c_{\chi^2} = 1/2$, so for any given D_ϕ it admits the least Exploitability, which is exactly the property that wins at high α .

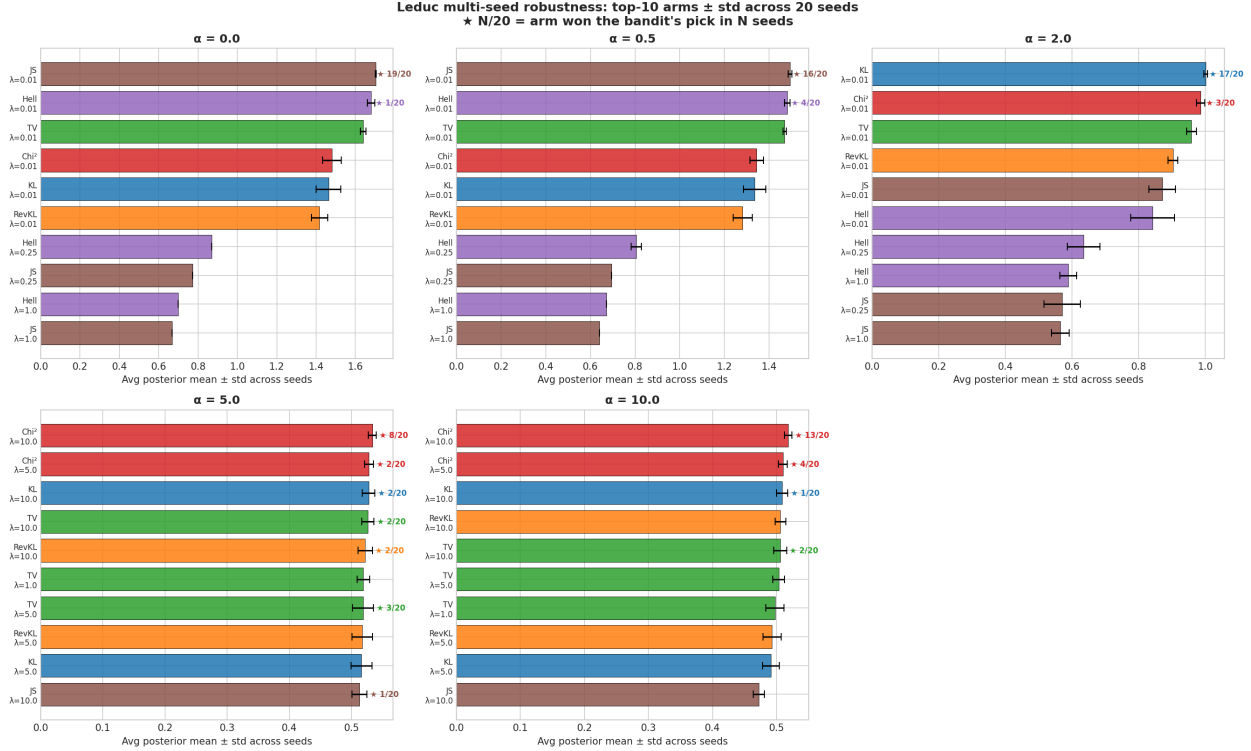


Figure 4: Top-10 arms ranked by average posterior mean across 20 seeds. Error bars show one standard deviation.

Figure 4 shows the average posterior mean per arm regardless of whether it was the best-arm selection. The spread between the top arms is wide at low α (the JS-Hellinger gap is roughly 0.05 at $\alpha = 0$) but compresses at high α . At $\alpha = 10$, the top three arms are within 0.006 of one another. This compression reflects the convergence of all divergences to π_i^{NE} as λ grows, after which the regularized policies have nearly identical rewards. The standard deviations of posterior means across seeds are small relative to the gaps between arms at low and moderate α , indicating that the bandit's ranking in those regimes is highly reproducible across exploration randomness.

In summary, the Leduc bandit identifies three regimes for optimal (ϕ, λ) selection. In the EV-focused regime ($\alpha \leq 0.5$), Jensen-Shannon at $\lambda = 0.01$ wins. In the balanced regime ($\alpha = 2$), Forward KL at $\lambda = 0.01$ wins, trading top-line Expected Value for tighter Exploitability while keeping the regularizer light. In the safety-dominated regime ($\alpha \geq 5$), Chi-Squared at large λ wins by driving Exploitability closer to zero than any other divergence. No single (ϕ, λ) dominates the entire range of α ; the right choice depends on the player's risk tolerance, and the bandit recovers this choice reliably across the regimes.

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